

PEDLAS Framework v1.0 (Draft)

Probability Bias, Elimination Threshold, Diversity Constraint, Leg Count, Anchor Distance, Slip Separation

Subtitle: A Structural Outcome-Space Compression Framework for Binary Football Betting Markets.

Abstract

PEDLAS is a structural optimization framework designed to compress a binary outcome space into an affordable set of candidate betting slips. It is not a prediction model and does not claim to overcome bookmaker margin or guarantee profit.

1. Motivation

With L eligible binary markets there are 2^L possible Under/Over outcome combinations. For $L=39$ this exceeds 549 billion possibilities, making exhaustive coverage impossible.

2. Market Selection

PEDLAS selects binary total-goal markets where the Under price is at least 1.20. This serves two purposes: (1) every qualifying leg is eligible for Betway Win Boost; (2) the selected goal line (e.g. Under 4.5, 5.5, 6.5) covers many realistic scorelines while retaining meaningful burst probability.

3. Binary Representation

Each leg is represented as U (Under) or O (Over). A betting slip is therefore a binary string of length L .

4. Core Parameters

P – Probability Bias

Expected dominance of Under selections. Adaptive, not fixed.

E – Elimination Threshold

Rejects structurally implausible runs of identical outcomes.

D – Diversity Constraint

Limits representation from any one competition.

L – Leg Count

Defines the search-space size and Win Boost tier.

A – Anchor Distance

Minimum Hamming distance from the anchor (typically all-Under) slip.

S – Slip Separation

Ensures generated slips are sufficiently different to avoid budget waste.

5. Compression

Compression Ratio (CR) = $2^L / N$, where N is the number of generated candidate slips after PEDLAS filtering. Higher CR indicates greater reduction of the original search space.

6. Budget Layer (External)

Budget is intentionally not a PEDLAS parameter. After PEDLAS generates N candidate slips, the Budget Optimizer computes $K = \text{floor}(\text{Budget}/\text{Stake})$ and selects the highest-ranked K slips.

7. Optional Ranking Layer

Candidate slips may be ranked using odds-implied uncertainty, historical league burst rates, goal-line height, and fixture characteristics. Ranking is optional and separate from PEDLAS.

8. Algorithm Pipeline

Input Pool -> Binary Encoding -> Apply P -> Apply E -> Apply D -> Apply A -> Apply S -> Candidate Slip Set -> Budget Optimizer -> Place K Slips

9. What PEDLAS Does

Compresses the outcome space, enforces structural realism, encourages diversity, reduces redundant slips, and produces a budget-ready candidate set.

10. What PEDLAS Does Not Do

PEDLAS does not predict football outcomes, increase the probability of a single bet winning, eliminate bookmaker margin, or guarantee profitability.

11. Future Work

Adaptive parameter estimation, Monte Carlo simulation, Bayesian updating, machine-learning-assisted ranking, and extensive historical backtesting.

Conclusion

PEDLAS should be viewed as a combinatorial optimization framework for generating diverse, structurally plausible coverage sets under a practical budget constraint.